

Classifiers and Kinds: Definitions and Rules

1 Classifier Structure

Classifiers form a tree structure rooted at $\top$:

$$c ::= \top \mid c.n \quad (\text{classifiers})$$

where $c.n$ denotes the n -th child of classifier c .

2 Kind Structure

Kinds represent sets of classifiers with possible exclusions:

$$K ::= \emptyset \mid K \cup K \mid c \setminus [c_1, \dots, c_n] \quad (\text{kinds})$$

The notation $c \setminus [\bar{c}]$ represents all classifiers in the subtree rooted at c , excluding the subtrees rooted at each c_i .

Abbreviations:

- $\text{top} \triangleq \top \setminus []$ (all classifiers)
- $\text{classifier}(c) \triangleq c \setminus []$ (subtree rooted at c)

3 Classifier Subclassing

The subclassing relation $c_1 \leq c_2$ holds when c_1 is in the subtree rooted at c_2 :

$$\frac{}{c \leq c} \text{SC-REFL}$$

$$\frac{c_1 \leq c_2}{c_1.n \leq c_2} \text{SC-PARENT}$$

4 Classifier Disjointness

Two classifiers are disjoint ($c_1 \perp c_2$) when they are in different subtrees:

$$\frac{n \neq m}{p.n \perp p.m} \text{CD-BASE}$$

$$\frac{c_1 \perp c_2}{c_1.n \perp c_2} \text{ CD-LEFT}$$

$$\frac{c_1 \perp c_2}{c_1 \perp c_2.m} \text{ CD-RIGHT}$$

5 Kind Disjointness

Two kinds are disjoint ($K_1 \perp K_2$) when they have no common classifiers:

$$\frac{}{\emptyset \perp K} \text{ KD-EMPTYL}$$

$$\frac{}{K \perp \emptyset} \text{ KD-EMPTYR}$$

$$\frac{K_1 \perp K \quad K_2 \perp K}{K_1 \cup K_2 \perp K} \text{ KD-UNIONL}$$

$$\frac{K \perp K_1 \quad K \perp K_2}{K \perp K_1 \cup K_2} \text{ KD-UNIONR}$$

$$\frac{r_1 \in \bar{e}_1}{r_1 \setminus \bar{e}_1 \perp r_2 \setminus \bar{e}_2} \text{ KD-ABSURDL}$$

$$\frac{r_2 \in \bar{e}_2}{r_1 \setminus \bar{e}_1 \perp r_2 \setminus \bar{e}_2} \text{ KD-ABSDRL}$$

$$\frac{r_1 \perp r_2}{r_1 \setminus \bar{e}_1 \perp r_2 \setminus \bar{e}_2} \text{ KD-ROOT}$$

$$\frac{r_1 \in \bar{e}_2}{r_1 \setminus \bar{e}_1 \perp r_2 \setminus \bar{e}_2} \text{ KD-EXCLL}$$

$$\frac{r_2 \in \bar{e}_1}{r_1 \setminus \bar{e}_1 \perp r_2 \setminus \bar{e}_2} \text{ KD-EXCLR}$$

Here, $r \in \bar{e}$ means $\exists e_i \in \bar{e}. r \leq e_i$ (containment via superclass).

6 Kind Intersection

The intersection relation $K_1 \cap K_2 = R$ computes the common classifiers:

$$\frac{}{\emptyset \cap K = \emptyset} \text{INT-EMPTYL}$$

$$\frac{}{K \cap \emptyset = \emptyset} \text{INT-EMPTYR}$$

$$\frac{K_1 \cap K = R_1 \quad K_2 \cap K = R_2}{(K_1 \cup K_2) \cap K = R_1 \cup R_2} \text{INT-UNIONL}$$

$$\frac{K \cap K_1 = R_1 \quad K \cap K_2 = R_2}{K \cap (K_1 \cup K_2) = R_1 \cup R_2} \text{INT-UNIONR}$$

$$\frac{r_1 \leq r_2}{(r_1 \setminus \overline{e_1}) \cap (r_2 \setminus \overline{e_2}) = r_1 \setminus (\overline{e_1} \uparrow \overline{e_2})} \text{INT-SUBL}$$

$$\frac{r_2 \leq r_1}{(r_1 \setminus \overline{e_1}) \cap (r_2 \setminus \overline{e_2}) = r_2 \setminus (\overline{e_1} \uparrow \overline{e_2})} \text{INT-SUBR}$$

$$\frac{r_1 \perp r_2}{(r_1 \setminus \overline{e_1}) \cap (r_2 \setminus \overline{e_2}) = \emptyset} \text{INT-DISJ}$$

7 Kind Subtraction

The subtraction relation $K_1 - K_2 = R$ computes the classifiers in K_1 but not in K_2 :

$$\frac{}{\emptyset - K = \emptyset} \text{SUB-EMPTYL}$$

$$\frac{K_1 - K = R_1 \quad K_2 - K = R_2}{(K_1 \cup K_2) - K = R_1 \cup R_2} \text{SUB-UNIONL}$$

$$\frac{}{(r_1 \setminus \overline{e_1}) - \emptyset = r_1 \setminus \overline{e_1}} \text{SUB-EMPTYR}$$

$$\frac{(r_1 \setminus \overline{e_1}) - K_1 = R_1 \quad R_1 - K_2 = R_2}{(r_1 \setminus \overline{e_1}) - (K_1 \cup K_2) = R_2} \text{SUB-UNIONR}$$

$$\frac{}{(r_1 \setminus \overline{e_1}) - (r_2 \setminus \emptyset) = r_1 \setminus (r_2 \uparrow \overline{e_1})} \text{SUB-TREE}$$

Exclusion handling: For $(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2))$, using the identity $A - (B - C) = (A - B) \cup (A \cap B \cap C)$:

$$\frac{a < r_2}{(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2)) = r_1 \setminus \bar{e}_1} \text{SUB-EXCLABSURD}$$

$$\frac{a \perp r_2 \quad (r_1 \setminus \bar{e}_1) - (r_2 \setminus \bar{e}_2) = R}{(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2)) = R} \text{SUB-EXCLIRRELR}$$

$$\frac{a \leq r_2 \quad a \leq r_1 \quad (r_1 \setminus \bar{e}_1) - (r_2 \setminus \bar{e}_2) = R}{(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2)) = R \cup (a \setminus \bar{e}_1)} \text{SUB-EXCLSUBR}$$

$$\frac{a \leq r_2 \quad r_1 < a}{(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2)) = r_1 \setminus \bar{e}_1} \text{SUB-EXCLSUBL}$$

$$\frac{a \leq r_2 \quad r_1 \perp a \quad (r_1 \setminus \bar{e}_1) - (r_2 \setminus \bar{e}_2) = R}{(r_1 \setminus \bar{e}_1) - (r_2 \setminus (a :: \bar{e}_2)) = R} \text{SUB-EXCLIRRELL}$$

Here, $a < r$ denotes strict subclassing ($a \leq r$ and $a \neq r$).

8 Subkinding

The subkinding relation $K_1 \subset K_2$ holds when every classifier in K_1 is also in K_2 :

$$\frac{K_1 - K_2 = R \quad \text{isEmpty}(R)}{K_1 \subset K_2} \text{SK-SUB}$$

where $\text{isEmpty}(K)$ is defined inductively:

$$\frac{}{\text{isEmpty}(\emptyset)} \text{ISEMPTY-EMPTY}$$

$$\frac{r \in \bar{e}}{\text{isEmpty}(r \setminus \bar{e})} \text{ISEMPTY-ABSURD}$$

$$\frac{\text{isEmpty}(K_1) \quad \text{isEmpty}(K_2)}{\text{isEmpty}(K_1 \cup K_2)} \text{ISEMPTY-UNION}$$